

LLM Agents Forecasting Euro Area Monetary Policy Transmission

System design of a pre-registered live forecasting experiment
Round 1: `ea-2026q3-live`

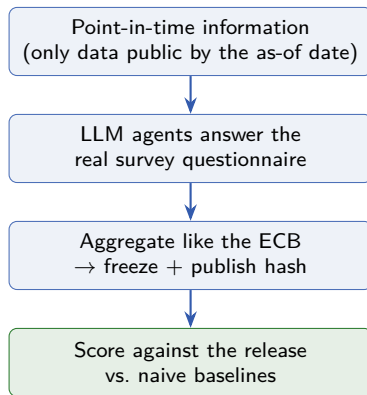
October 2026

The idea in one slide

Question. Can LLM agents that role-play euro area **banks** and **households** forecast how the real survey respondents reacted to the 2026 tightening — *before* the surveys are published?

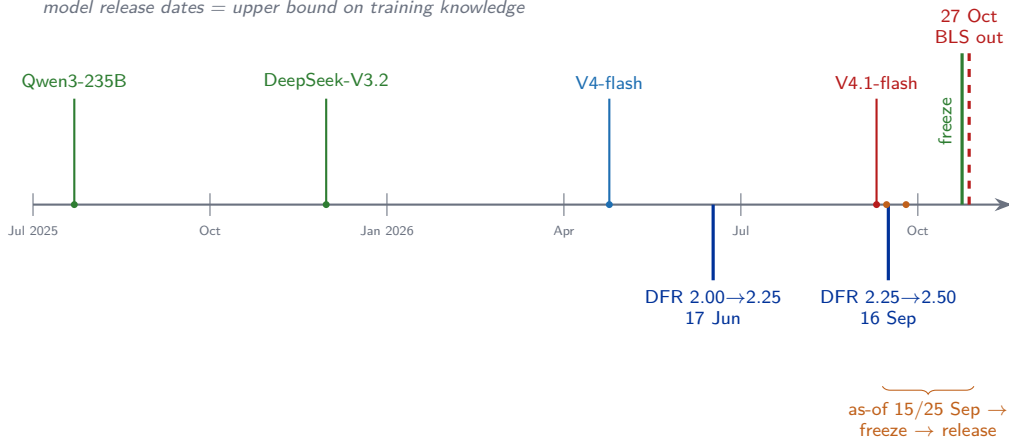
Why live? Back-testing on past surveys is contaminated: the answers may sit in the training data. Forecasting an *unreleased* number is the only clean test.

Second question. Does any skill come from *reasoning* or from *memorised* data? A panel of models released before and after the shock answers this.



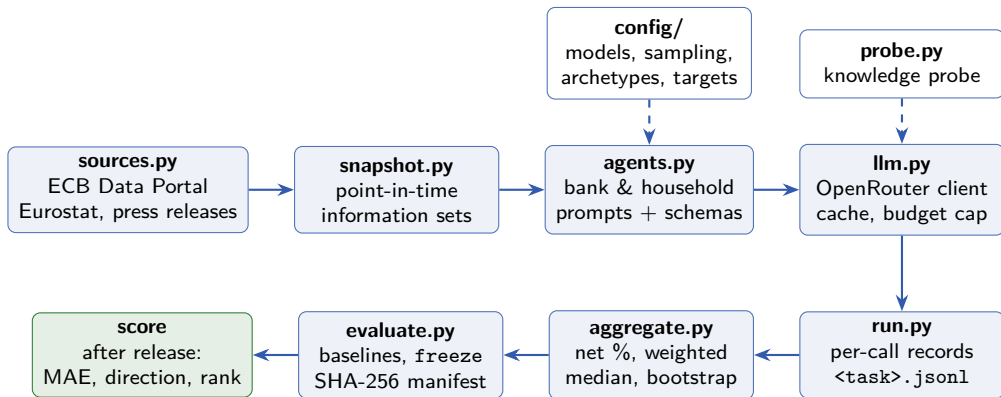
Timeline: shock, knowledge gradient, forecast window

model release dates = upper bound on training knowledge



Arms: **clean** (pre-shock) **pre-hike** **post-hike** (contamination risk). Freeze before 26 Oct; the CES September wave follows in late October.

System architecture



One CLI: data → estimate → run --tasks probe → run --tasks bls ces → baselines → freeze → score

Point-in-time information sets

Agents see only data whose *conservative* publication date is on or before the as-of date — and no narrative written by the researchers.

Task	As of	Contents
CES households	15 Sep 2026	own-country HICP; first paragraph of the 10 Sep ECB press release; own-country mortgage rate (mortgagors only)
BLS banks	25 Sep 2026	DFR history, €STR, 12m Euribor, HICP (EA, DE, FR, IT, ES), MFI lending rates, previous BLS rounds, full 10 Sep press release

HICP	→	available from day 5 of next month
MFI interest rates	→	available from day 6, two months later
BLS round	→	available from day 28 of first month of its SDW period
Daily rates	→	available from next day

Knowledge probe: does a model already know the 2026 hikes?

Six DFR questions, 3 samples each (1 at $T=0$, 2 at $T=1$):

- ▶ **control** (2024, 2025): can the model answer at all?
- ▶ **boundary** (1 Jan 2026): answerable by assuming no change
- ▶ **target** (Jun, Sep 2026, first hike): only if trained on post-shock data

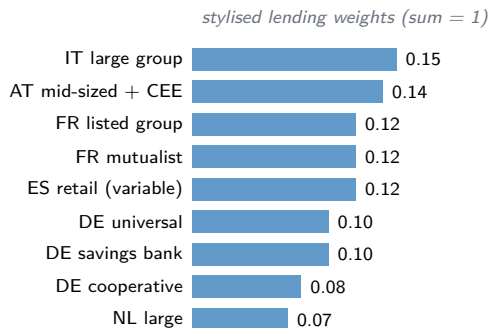
Flag “knows 2026” = any target correct.
Diagnostic only; nobody is excluded.

Round 1 result (correct answers out of 3)

	Qwen3	DS-V3.2	V4-flash	V4.1-flash
2024-06 control (3.75)	2	3	3	3
2025-06 control (2.00)	0	0	0	2
2026-01 boundary (2.00)	0	0	0	1
2026-06 target (2.25)	0	0	0	0
2026-09 target (2.50)	0	0	0	0
First hike (2.25)	0	0	0	0

No model knows the 2026 hikes. Knowledge fades through 2025 even for the September 2026 model.

Bank agents: nine archetypes answer the BLS questionnaire

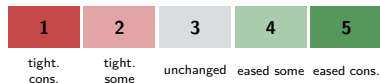


Prompt system: *"You are the senior loan officer . . . Today's date is 2026-09-25; you know nothing that happened after it."*

user: data block + ECB press release + six questions.

Questions credit standards for {enterprises, housing, consumer credit} $\times$ {past 3m, next 3m}

Answer scale (ECB coding)



Structured JSON output with a `main_factors` rationale.
30 samples per archetype per model at $T=1$.

Household agents: 60 personas, two information arms

	Q1	Q2	Q3	Q4	Q5
18–34	R	R	R	M	M
35–54	R	R	M	M	M
55–70	R	O	O	O	O

R renter M mortgagor O outright owner
× 4 countries: DE, FR, IT, ES = 60 personas

Persona = age band (weights .27/.38/.35) × income quintile (equal); mortgagors hold the country's dominant mortgage type (ES variable, DE/FR fixed, ...).

Two arms

- ▶ **news** (primary): own-country HICP, the ECB announcement, own mortgage rate — facts a household could have seen.
- ▶ **minimal**: persona and date only. A model that still produces 2026-level answers here betrays memorised data.

Output: perceived past and expected next-12-month inflation (%). 8 samples per persona × arm × model.

Aggregation: from agent answers to survey statistics

BLS net percentage

$$\text{Net}_q = 100 \sum_b w_b \left(\underbrace{\text{Pr}_b[1-2]}_{\text{tightened}} - \underbrace{\text{Pr}_b[4-5]}_{\text{eased}} \right)$$

for each of the 6 questions q ; shares are over bank b 's 30 samples. Positive = net tightening, as in the ECB release.

CES country median

$$\text{Med}_c = \text{wmedian} \left\{ x_{p,s}; \frac{w_p}{n_p} \right\}_{p \in c}$$

weighted median of `expected_inflation_next_12m` over all persona samples in country c . Answers with $|x| > 100$ are dropped and counted.

Uncertainty and failures

90% intervals from 1,000 bootstrap draws resampling answers *within* each bank / persona. Failed calls (API or JSON errors) are counted and reported, never imputed.

Targets, baselines and hypotheses

Targets (ECB Data Portal)

Survey	Statistic	Release
BLS	3 items $\times$ past/next 3m	27 Oct 2026
CES	median 12m exp., DE/FR/IT/ES	late Oct 2026

Baselines (fetched before freezing)

- ▶ BLS persistence: previous round's value
- ▶ BLS banks' own previous expectation (past-3m targets)
- ▶ CES persistence: August 2026 value

Hypotheses

- H1 Bank agents beat both BLS baselines on MAE.
- H2 Household agents (news) beat CES persistence on MAE and rank the four countries correctly.
- H3 Skill does *not* rise with model release date; pre-shock models give no 2026-level answers in the minimal arm.

Metrics: MAE (pp), direction hits vs. previous value, Spearman rank (CES). Persistence is a strong baseline — a null result is reported the same way.

Integrity safeguards

No look-ahead

point-in-time data; “you know nothing after the as-of date”

Pinned models

dated versions only; served model id logged per call

Reproducible calls

request hash + response cache; per-call JSONL records

Hard budget cap

worst-case cost reserved before every call

Freeze + public hash

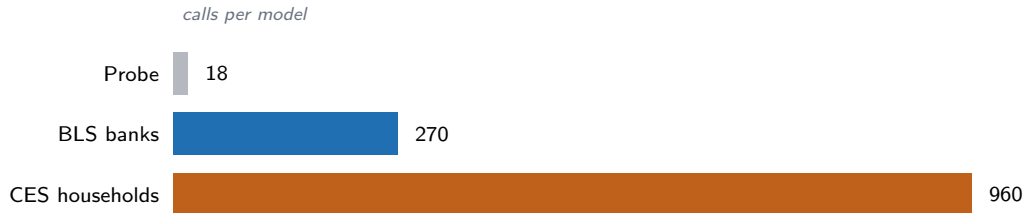
SHA-256 of predictions.json posted (OSF)

Freeze refuses

if any target is already published

Pre-registered design; every post-freeze change is logged under *Deviations*.

Scale of round 1



BLS: 9 archetypes × 30 samples	270
CES: 60 personas × 2 arms × 8 samples	960
× 4 models ⇒ forecast calls	4,920
Worst-case cost (budget cap \$10 per CLI call)	\$8.40

Roadmap

